

aim:

to calculate the Subtransient Current in the faulted phase when unsymmetrical fault takes place.

objectives:

1. to become familiar with modeling and analysis of power system under faulted condition and to compute the fault level, post fault voltage and currents for different types of faults, both symmetrical and unsymmetrical.
2. to calculate the fault current, post fault voltage and fault current through the branch for a 3 ϕ to ground fault in a small power system and also study the effect of neighbouring system. check the results using available software.

Software Required:

MATLAB

Formula:

1) Single line to ground fault:

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_3}$$

$$I_b = \frac{KV_b^2}{MVA_b}$$

$$\text{iii) } I_{\text{fault}} = I_b * I_F$$

$$\text{iv) } I_b = \frac{KV_{ab}}{KV_b^{1/3}}$$

Line to Line Fault:

$$\text{i) } I_{a1} = \frac{E_a}{Z_1 + Z_2}$$

$$\text{ii) } I_a = I_{a0} + I_{a1} + I_{a2}$$

$$\text{iii) } V_{a0} = 0; \quad V_{a1} = E_a - I_{a1} Z_1; \quad V_{a2} = V_{a1}$$

$$\text{iv) } V_a = V_{a0} + V_{a1} + V_{a2}; \quad V_b = V_{a0} + a^2 V_{a1} + a V_{a2}; \quad V_c = V_b$$

$$\text{v) } V_{ab} = V_a - V_b; \quad V_{bc} = V_b - V_c; \quad V_{ca} = V_c - V_a;$$

$$\text{Phase voltage or base voltage} = (KV_b)^{1/3}$$

iii) Double line to ground Fault:

$$\text{i) } I_{a1}, I_a = I_{a0}$$

$$\text{ii) } I_{a1} = \frac{E_a}{Z_1 + \frac{Z_2(Z_0 + 3Z_2)}{Z_2 + Z_0 + 3Z_2}}$$

$$\text{iv) } V_{a1} = V_{a0} = \frac{E_a - I_{a1} Z_1}{\frac{Z_2 + Z_0 + 3Z_2}{Z_2 + Z_0 + 3Z_2}}$$

Procedure:

1. open matlab -> Simulink -> File -> New -> Model
2. open Simulink Library and browse the components
3. Connect the components as per circuit diagram
4. set the desired voltage and sequence frequency

Result:

thus the subtransient current in the faulted phase under unsymmetrical fault condition was analysed & verified